

Personal Search Plan

Research question

How to find the cause of the unfeasibility of the logic of Feature Models?

Example options for broadening or narrowing the research topic:

- Focus on the specific implementation of Feature Models as done in the company Elfsquad.

Databases

SmartCat; IEEE Xplore; Scopus (Elsevier); Google scholar

Known items

These are references that you want to use and already know about

“Knowledge-based configuration; From research to business cases.”

- Alexander Felfernig, Lothar Hotz, Claire Bagley, and Julia Tiihonen.

A detailed book about the Constraint Satisfaction Problem’s specific implementation of Feature models.

“unsatisfiability of Feature Models”

The compact definition of the problem I’m trying to research.

Search terms

Constraint Satisfaction Problem
Feature Models
Constraints
Cross-Tree Constraints
Binary Constraints
Nary Constraints
Expressions
Diagnostics
Unsatisfiability
Theorem Proof generator
Inference engines
First order logic
Higher order logic

Search string

Basic

kw:(Constraint Satisfaction Problem) AND kw:(unsatisfiability) (42 results in University of Groningen)

Summary of sources found

This section can contain examples of relevant findings or a table of results if many results are found in different databases

Book:

“Knowledge-based configuration: from research to business cases”.

This book gives a lot of the relevant background and currently implemented functionality that exist within Elfsquad. After working with this for some time, we figured out that this book alone is not exactly enough for what we wanted to do.

This meant we have extended upon the generally accepted definition of Feature Models.

For this extension however, there was nearly no research to be found. Mostly general ideas about diagnosing a Feature Model that were not relevant for our implementation, meaning that it was mostly of no use.

Preliminary conclusions on the search results and the search process

Your thoughts on how it went

As expected, not much research can be found on the specific implementation for which I am searching. This is because I am not aware of any company/research delving into the specific implementation that is being used at Elfsquad.

This is mostly due to the fact that this is considered private business matters of course.